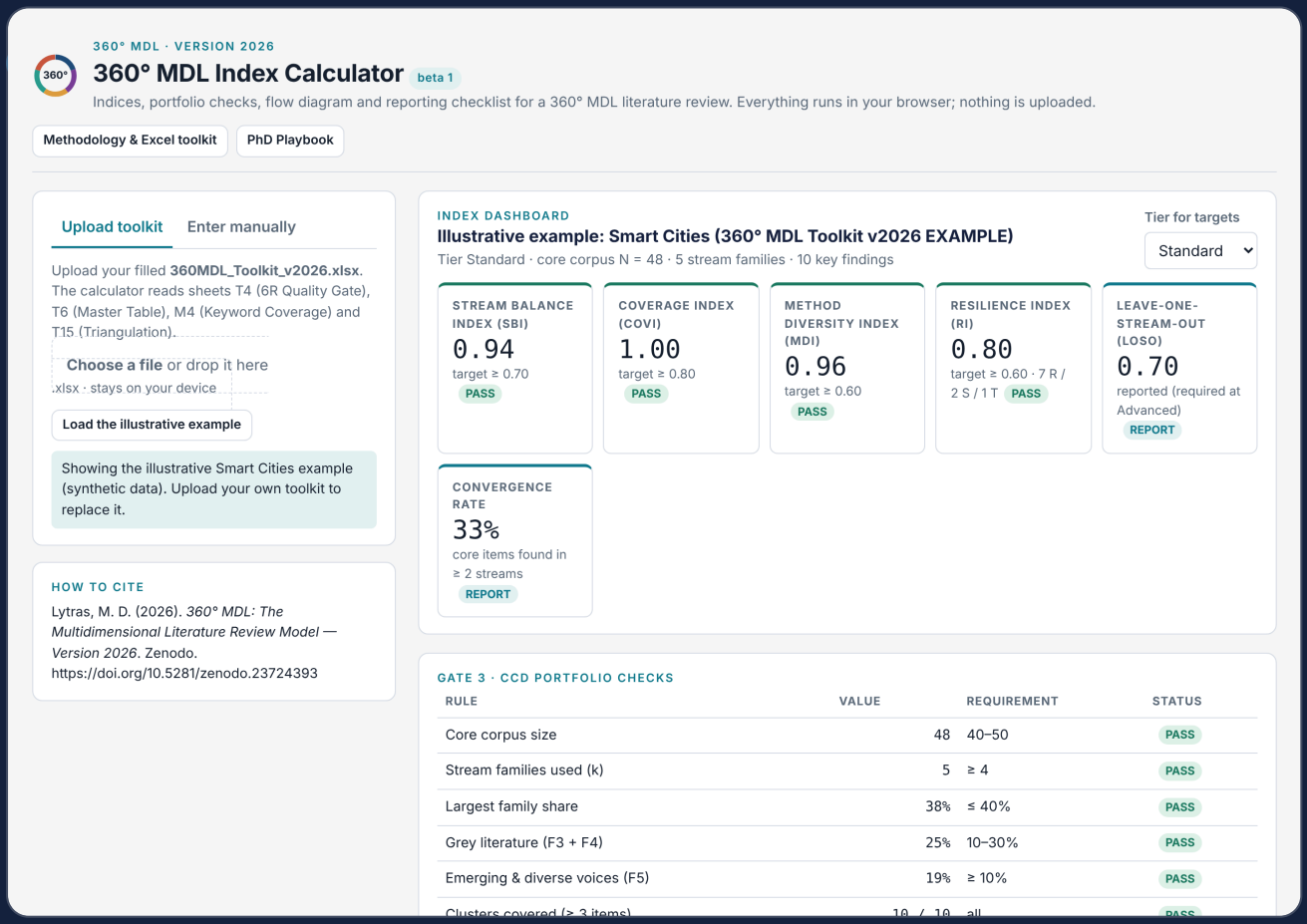
360° MDL · VERSION 2026 · OPEN TOOL · CC BY 4.0 / MIT

USER GUIDE & RESPONSIBLE-AI PROTOCOL

360° MDL Index Calculator

Compute the indices, check the portfolio, draw the flow diagram and export publication-ready figures for a 360° MDL literature review, in your browser, with nothing uploaded. Includes the Claude protocol for responsible AI assistance.



Miltiadis D. Lytras
Deree — The American University of Greece · Effat University
Beta 1 · November 2026

THE 360° MDL ECOSYSTEM

Where this tool sits

Resource	What for	Where
Methodology, handbook, quick start, Excel toolkit	The method and its templates T1–T17	https://doi.org/10.5281/zenodo.23724393
PhD Literature Review Playbook (EN / GR)	Step-by-step 12-week route for doctoral researchers	https://doi.org/10.5281/zenodo.23777000 · https://doi.org/10.5281/zenodo.23807964
Index Calculator (this tool)	Indices, checks, diagram, figures, checklist	https://doi.org/10.5281/zenodo.24033376
Applied reviews and open data	Two full demonstrations with record-level data	https://doi.org/10.5281/zenodo.23772941
Community	Deposit your 360° MDL review or protocol	https://zenodo.org/communities/360mdl

360° MDL Index Calculator — User Guide and Responsible-AI Protocol. Beta 1, November 2026.

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How to cite the tool: Lytras, M. D. (2026). *360° MDL Index Calculator* (Version beta 1) [Software]. Zenodo. https://doi.org/10.5281/zenodo.24033376

How to cite the methodology: Lytras, M. D. (2026). *360° MDL: The Multidimensional Literature Review Model — Version 2026*. Zenodo. https://doi.org/10.5281/zenodo.23724393

Licences: the software (HTML/JavaScript) is released under the MIT Licence; this guide and the figures it produces are released under Creative Commons Attribution 4.0 International (CC BY 4.0). © 2026 Miltiadis D. Lytras. 360° MDL is a trademark of the author.

Privacy: the calculator runs entirely in the browser. No file, number or text is transmitted to any server.

Use of AI: generative AI (Claude, Anthropic) assisted with the development of the software and the drafting of this guide under the author's direction. The author reviewed and approved all content.

CHAPTER 1

What the calculator does

The 360° MDL Index Calculator turns the numbers of your review into the outputs a 360° MDL report requires: indices, portfolio checks, flow diagram, triangulation table, reporting checklist and nine publication-ready figures and tables.

Inputs

- The filled **360MDL_Toolkit_v2026.xlsx** (sheets T1, T4, T6, M4 and T15 are read), or
- **Manual entry**: core items per stream family, method families, clusters, gate counts, convergence and key findings.
- The **tier** (Lite, Standard, Advanced), which sets the targets.

Outputs

- Index dashboard: SBI, CovI, MDI, RI, LOSO, Convergence Rate, with PASS / CHECK against the targets.
- Gate 3 CCD portfolio checks (eight rules).
- PRISMA 2020-compatible flow diagram.
- Triangulation and resilience table (T15) with LOSO per family.
- MDL-RC 2026 checklist with computed items.
- Six figures and three tables as PNG (2×), tables also as text for Word or Excel.
- A ready methods sentence, a text report and a JSON record.

THREE WAYS TO RUN IT

1. **Online.** Open the public link (Chapter 2). Nothing to install.
2. **Offline.** Download the single HTML file and double-click it. It works without internet after the first load of its fonts.
3. **On your university site.** Upload the HTML file to any web server or GitHub Pages; it needs no backend.

WHO IT IS FOR

Doctoral researchers writing a 360° MDL chapter, review authors preparing a journal submission, supervisors checking a student's corpus, and editors or reviewers auditing a submitted 360° MDL review.

WHAT IT DOES NOT DO

It does not search databases, score items or write findings for you. Those are judgement tasks that the methodology keeps with the researcher. The calculator computes, checks and draws.

CHAPTER 2

Get the calculator

Option A · Online (nothing to install)

1. Open the link: `https://mlytras.github.io/360mdl/calculator.html`
2. The page opens with an illustrative example already loaded, so you can see every output before adding your own data.
3. Bookmark it. Your data never leaves your browser.

Option B · Offline, on your computer

1. Download **360MDL_Index_Calculator_beta1.zip** from Zenodo (https://doi.org/10.5281/zenodo.24033376).
2. Unzip it. You will find **360MDL_Index_Calculator.html**, this guide, a README and the licence.
3. Double-click the HTML file. It opens in your default browser (Chrome, Edge, Firefox or Safari, any recent version).
4. The first time, the browser fetches two small resources (the fonts and the Excel reader library). After that the page also works without internet; if offline, the page still works but uses your system font.

Option C · Host it for your group

1. Upload the HTML file to any web server, a university page or a free GitHub Pages site.
2. No database, no backend and no configuration are required.
3. Keep the file name and the version note in the footer, so users know which release they run.

SYSTEM REQUIREMENTS

- Any modern browser (2022 or later).
- JavaScript enabled.
- For uploading toolkits: an .xlsx file saved by Excel, LibreOffice or Google Sheets.
- Works on phones and tablets; the figures export is best on a desktop.

SECURITY NOTE

The calculator reads your workbook with a JavaScript library inside the browser. It never sends the file anywhere. If your institution's policy forbids opening local HTML files, use the online link or ask your IT service to host the file.

UPDATES

New versions are published on Zenodo under the same concept DOI, so the citation stays valid. The footer of the page shows the version you are running.

CHAPTER 3

Step 1 · Load your data

Upload the filled toolkit

1. Complete your **360MDL_Toolkit_v2026.xlsx** at least up to sheet T15 (key findings). The more sheets you fill, the more the calculator can compute.
2. Click **Choose a file** or drag the workbook onto the dotted box.
3. The status line confirms how many screened items, core items and key findings were read.
4. Check the **Tier for targets** selector on the right; it is pre-filled from sheet T1.

What the calculator reads

Sheet	Used for
T1 Review Canvas	Review title and tier
T4 6R Quality Gate	Screened items, scores R1–R6, number of streams each item was found in (Gate 2 outcomes and Convergence Flags)
T6 Master Table	The core corpus: IDs, method family, lead author or group
M4 Keyword Coverage	Clusters and the “x” marks per item (Coverage Index)
T15 Triangulation	Key findings and the stream families that support them

The stream family of every item is derived from its ID prefix (WOS, SCO, GSC, DOM → F1; HCR, T2P, GSP, RCO, ADS → F2; IND → F3; GOV → F4; EMV → F5). Keep the ID scheme `[STREAM]-[SLICE]-[NN]` exactly as in the toolkit.

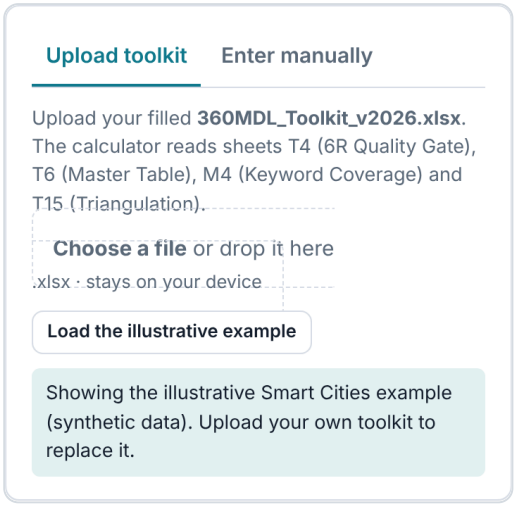


Figure 1. The upload panel. The illustrative example can be loaded at any time to see a complete output.

COMMON PROBLEMS

- **“Sheets T4 and T6 were not found.”** The file is not the 360° MDL toolkit, or the sheet names were renamed.
- **“No items found.”** The ID column (B) is empty. Fill IDs such as SCO-IMP-01 in T4 and T6.
- **Families all F1.** Check the ID prefixes.
- **No key findings.** Mark the “x” cells in T15 or type the findings manually.

CHAPTER 3 · CONTINUED

Or enter the numbers manually

If you work without the Excel toolkit, for example at the Lite tier with a simple table, switch to **Enter manually** and type the figures you already have.

1. **Tier.** Lite, Standard or Advanced; sets the targets.
2. **Core corpus items per stream family.** How many core items came from F1 … F5.
3. **Core items per method family.** The nine families of matrix M2: Quantitative, Qualitative, Mixed, Design science/Artefact, Conceptual/Theoretical, Review/Bibliometric, Simulation/Modelling, Case study, Delphi/Expert.
4. **Keyword clusters.** The total number of clusters and how many are covered by at least three core items (Lite: two).
5. **Gate counts.** Records harvested, unique after de-duplication, eligible after Gate 1; Gate 2 pass, reserve, exclude. These feed the flow diagram.
6. **Convergence.** Core items that were found in two or more streams.
7. **Key findings.** One row per finding, with the stream families that support it ticked. Add or remove rows as needed.
8. Click **Compute**.

> **TIP**
>
> Start by loading the illustrative example, then switch to manual entry: the form is pre-filled with plausible values that you can overwrite one by one.

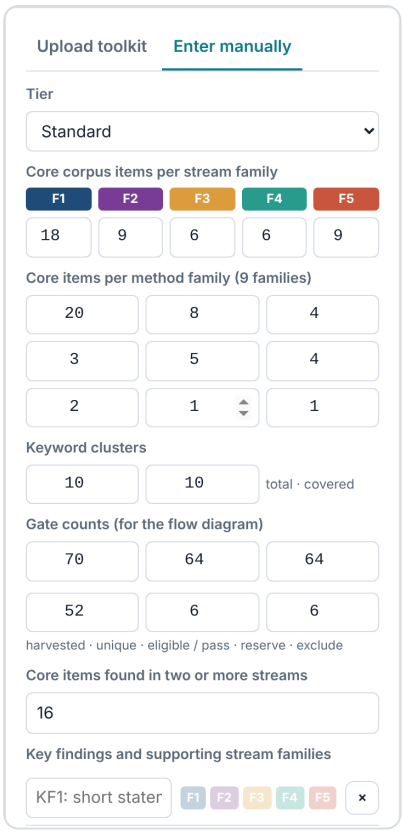


Figure 2. Manual entry (upper part). Family chips next to each key finding are toggled by clicking.

CHAPTER 4

Step 2 · Read the index dashboard

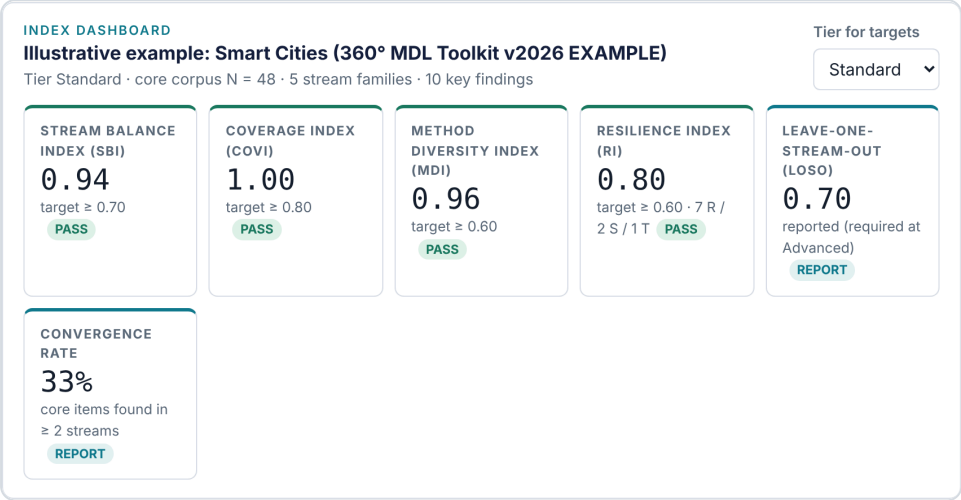


Figure 3. The index dashboard with the illustrative example at the Standard tier.

Index	Definition (specification §7)	Target	Reading
SBI Stream Balance Index	Normalised Shannon entropy of the core corpus over the k stream families used: H / ln k	≥ 0.70	1.00 = perfectly balanced; low values mean one family dominates
CovI Coverage Index	Share of keyword clusters covered by ≥ 3 core items (Lite ≥ 2)	≥ 0.80	Clusters below the threshold are white spots
MDI Method Diversity Index	Normalised Shannon entropy over the nine method families: H / ln 9	≥ 0.60	A low MDI can be a finding about the field, not a defect of the review
TS Triangulation Strength	Number of stream families supporting a key finding	Robust ≥ 3	Supported = 2, Tentative = 1
RI Resilience Index	(#Robust + 0.5 × #Supported) / #key findings	≥ 0.60	How much of the synthesis rests on triangulated findings
LOSO Leave-One-Stream-Out	Share of findings that keep TS ≥ 2 when one family is removed; worst case over families	≥ 0.70 (Advanced)	Reported at all tiers; shows which family the review depends on
Convergence Rate	Share of core items found in ≥ 2 streams	reported	High values can signal a tightly connected research network

PASS, CHECK AND REPORT

PASS: the index meets the target of the selected tier. **CHECK**: it does not; either improve the corpus (for example add F5 items) or report the shortfall with a justification (MDL-RC item 30). **REPORT**: the index has no target at this tier and is simply reported.

CHAPTER 4 · CONTINUED

Portfolio checks and flow diagram

GATE 3 · CCD PORTFOLIO CHECKS

RULE	VALUE	REQUIREMENT	STATUS
Core corpus size	48	40–50	PASS
Stream families used (k)	5	≥ 4	PASS
Largest family share	38%	≤ 40%	PASS
Grey literature (F3 + F4)	25%	10–30%	PASS
Emerging & diverse voices (F5)	19%	≥ 10%	PASS
Clusters covered (≥ 3 items)	10 / 10	all	PASS
Method families represented	9	≥ 3	PASS
Max items per lead author/group	2	≤ 3	PASS

Figure 4. Gate 3 CCD checks. A rule marked JUSTIFY needs a written justification in the review.

The eight rules of Gate 3: core corpus size within the tier range; at least the required number of stream families; no family above 40%; grey literature (F3 + F4) between 10% and 30%; F5 at least 10% (Advanced 15%); every cluster covered; at least three method families; no lead author or group with more than three items. The last two rules need sheets T6 and M4; with manual entry they show n/a.



Figure 5. The flow diagram, compatible with PRISMA 2020. Identification by stream family, then the Three Gates.

The numbers come from T4 (screened items and Gate 2 outcomes) and T6 (core corpus). Database hit counts before export belong in your Stream Log (T3) and are reported in the methods section; the diagram starts at the records you screened.

CHAPTER 4 · CONTINUED

Triangulation, checklist and report

TRIANGULATION & RESILIENCE · T15

KF	KEY FINDING	SUPPORTING FAMILIES	TS	CLASS
KF1	Governance and leadership capacity is a precondition for smart-city transformation.	F1 F2 F4 F5	4	Robust
KF2	Structured citizen participation increases legitimacy and uptake of smart-city services.	F1 F4 F5	3	Robust
KF3	Open data generates value only when combined with data governance and interoperability.	F1 F3 F4	3	Robust
KF4	Interoperability of IoT infrastructure is the main barrier to scaling pilots.	F1 F2 F3	3	Robust
KF5	Sustainability outcomes are rarely measured with consistent, comparable indicators.	F1 F3 F4 F5	4	Robust
KF6	Digital twins and AI are promising capabilities, but evidence remains mostly pilot-level.	F1 F2 F3 F5	4	Robust
KF7	Trust, privacy and security are decisive conditions for adoption.	F1 F2 F4	3	Robust
KF8	Data-driven mobility services produce measurable efficiency gains.	F1 F3	2	Supported
KF9	The economic value of smart energy depends on regulation and user behaviour.	F3 F4	2	Supported
KF10	Global South cities pursue distinct leapfrogging pathways.	F5	1	Tentative

LOSO: SHARE OF FINDINGS KEEPING TS ≥ 2 WITHOUT EACH FAMILY

worst case = 0.70

F1	F2	F3	F4	F5
0.80	0.90	0.70	0.80	0.90

Figure 6. The triangulation table. Coloured chips show the supporting families; faded chips show the missing ones. The footer gives LOSO without each family.

The MDL-RC 2026 checklist

Items 8, 14, 15, 20, 21, 22 and 23 are filled from your data and marked **computed**, or **below target** when an index misses its target. Tick the other items as you report them in the manuscript. The counter shows how many of the 30 items are covered. Submit the checklist with your review.

The report

The **methods sentence** is ready to paste into your manuscript. **Copy full report** gives a plain-text record of every number and check, useful as a supplementary file. **Copy JSON** gives a machine-readable record for your data deposit. Nothing is saved on the page itself: keep your workbook as the source of truth.

EXAMPLE METHODS SENTENCE (ILLUSTRATIVE DATA)

This review was conducted and reported according to the 360° MDL methodology, Version 2026 (Lytras, 2026), at the Standard tier, with 5 stream families and a core corpus of 48 items. Stream Balance Index 0.94, Coverage Index 1.00, Method Diversity Index 0.96; 7 of 10 key findings were Robust (Resilience Index 0.80; Leave-One-Stream-Out worst case 0.70).

CHAPTER 5

Step 3 · Export figures and tables for your paper



Figure 7. The export section: six figures and three tables generated from your data, each with a caption and a download button (first four shown).

File	Content
Figure_1_flow_diagram	PRISMA-compatible flow diagram
Figure_2_core_composition	Share of each stream family in the core corpus, with CCD limits
Figure_3_index_profile	Indices against tier targets
Figure_4_triangulation_matrix	Key findings × families, TS and class
Figure_5_LOSO	Stability without each family
Figure_6_method_diversity	Method families of the core corpus
Table_1_stream_analysis	Harvest, core items and share per family
Table_2_indices	Indices, targets, status, definitions
Table_3_CCD_checks	Portfolio rules and status

- Images are PNG at twice screen resolution, white background, suitable for journals at single-column width. For very large prints, redraw from the JSON.
- **Download PNG** asks for your confirmation when the tool runs inside claude.ai; in the offline file it saves directly. If a download does nothing, right-click the image and choose *Save image as…*.
- **Copy table for Word/Excel** copies the table as tab-separated text; paste it into Word (Insert → Table → Convert Text to Table) or Excel.
- Every figure carries the note “360° MDL v2026 · Index Calculator”; keep it, or cite the tool in the caption.

SUGGESTED CAPTION WORDING

“Figure generated with the 360° MDL Index Calculator (Lytras, 2026) from the review's toolkit data.”

CHAPTER 6

The Claude protocol: responsible AI with 360° MDL

Generative AI can save weeks in a literature review and can also invent references, flatten nuance and copy the biases of its training data. 360° MDL allows AI assistance under one condition, the **Two-Key Rule**: every AI output is verified by a human against the original source before it enters the review, and the use is disclosed.

The four keys of the rule

1. **Verified source.** No item enters the corpus unless a human has opened a page showing its title, authors or issuing body, year and DOI or official URL.
2. **Human scores.** AI may propose 6R scores and codes; a human confirms every one.
3. **Traceable text.** Every key finding cites item IDs; AI drafts are checked against those items.
4. **Disclosure.** The AI-use statement names the tool, the period, the tasks and the verification.

Allowed, assisted, forbidden

Task	Rule
Search-string ideation, synonyms	Allowed; humans choose the final strings
De-duplication, metadata cleaning	Allowed; spot-check
First-pass extraction to the Master Table	Allowed; verify every cell
Keyword clustering suggestions	Allowed; humans approve the codebook
Translation of regional scholarship	Allowed; disclose; check key passages
6R scoring, contribution coding	Assist only; human confirmation
Generating references, citing unread items	Forbidden
Uploading confidential or unpublished material	Forbidden

WHY CLAUDE

The protocol was developed and tested with Claude (Anthropic) in the two applied reviews of the methodology and in building this calculator. It works with any capable assistant; the prompts on the next pages are written for Claude and can be pasted into a Claude Project that holds your Review Canvas and toolkit.

SET UP A CLAUDE PROJECT FOR YOUR REVIEW

1. Create a Project named after your review.
2. Add the methodology overview, the quick-start guide and your Review Canvas (T1) as project files.
3. Paste the **system instruction** below into the project instructions.
4. Keep one chat per phase (P1 … P7).

PROJECT INSTRUCTION (PASTE AS IS)

```
You assist a 360° MDL literature review
(Lytras, 2026). Follow the Two-Key Rule:
never invent or guess references; every
source you propose must include a title,
authors or issuing body, year and a DOI or
official URL that I will verify; mark
anything unverified as [UNVERIFIED].
Propose 6R scores and codes only as
provisional. Use the ID scheme [STREAM]-
[SLICE]-[NN] and the toolkit sheets T1–
T17. Keep a log of your tasks for the AI-
use statement.
```

CHAPTER 6 · CONTINUED

Eight prompts, one per phase

Paste each prompt into your Claude Project with the material in brackets. Every prompt ends with the human step (“your key”) that the Two-Key Rule requires.

P1 Frame Review Canvas

prompt 1 of 8

```
I am starting a 360° MDL review on [TOPIC] at the [Lite/Standard] tier. Help me draft the Review Canvas (T1): a one-sentence purpose, 2–4 review questions from at least two families (what, why, how, what-if, should), the scope (years, languages, document types, geography), 6–10 provisional keyword clusters with synonyms, and inclusion/exclusion criteria written so two reviewers would apply them identically. Ask me questions where my topic is unclear.
```

Your key: You decide the final questions and criteria; check the clusters against 5–10 seed papers you have read.

P2 Source Search strings

prompt 2 of 8

```
Using the clusters in my Review Canvas, write Boolean search strings for Scopus (TITLE-ABS-KEY syntax), Web of Science (TS=) and Google Scholar, with truncation and phrase operators, for the Impact, Recency (last 2 years) and Frontier (special issues) slices. Then list the organisations and repositories I should search for F3 (industry) and F4 (policy) sources, and the sub-streams to try for F5. Do not list any papers.
```

Your key: Test every string yourself; your seed papers must appear. Log each run in T3 on the day you run it.

P2 Source Candidate discovery (with verification)

prompt 3 of 8

```
From the records I paste below (exported from [DATABASE]), identify which ones match my eligibility criteria and give for each: short title, lead author or body, year, DOI or official URL, candidate ID [STREAM]-[SLICE]-[NN], and a one-line reason. Mark any item whose details you cannot confirm from the pasted text as [UNVERIFIED]. Never add items that are not in the pasted records.
```

Your key: Open every DOI or URL yourself before entering the item in T4. Drop anything you cannot open.

P3 Select Provisional 6R scoring

prompt 4 of 8

```
For each item below (ID, title, abstract or summary, source type), propose provisional 6R scores (0–2) for R1 Relevance, R2 Rigour (use AACODS for grey literature), R3 Reach, R4 Recency, R5 Richness, R6 Representativeness, with a one-sentence rationale per criterion. Label every score PROVISIONAL.
```

Your key: Confirm or change each score in T4 after reading the full text; at Standard tier have a second coder score 20% and compute κ.

CHAPTER 6 · CONTINUED

Prompts 5–8

P4 Codify Master Table extraction

prompt 5 of 8

```
For item [ID] (full text pasted below), draft the Master Table (T6) fields: topics covered,
focus/context, methodology, method family (one of the nine), key findings (3–5), type of
contribution (CT1–CT8), significance, future research, limitations, key interpretation,
exploitation mode. Quote the passage that supports each field.
```

Your key: Verify each cell against the quoted passage; rewrite in your own words.

P4 Codify Keyword unification

prompt 6 of 8

```
Here are the author keywords and index terms of my core corpus. Normalise them (spelling,
singular/plural, acronyms, synonyms) and propose at most ten clusters with a short label and a
one-line definition each, and map every item ID to its clusters (≤ 8 per item). Show the
decisions you were unsure about.
```

Your key: Approve the codebook; adjust clusters so that each is covered by at least three items, or report the white spot.

P6 Synthesise Key findings

prompt 7 of 8

```
From the Master Table rows below, draft 6–12 key findings. Each finding must state a claim, its
scope, and the supporting item IDs. For each, list the stream families of the supporting items.
Flag findings supported by one family only as candidate Tentative findings and suggest what
evidence would strengthen them.
```

Your key: Check every ID; enter the findings in T15; run the calculator for TS, RI and LOSO.

P7 Report Methods section and AI statement

prompt 8 of 8

```
Using the calculator report pasted below, draft the methods section of my review in the 360° MDL
structure (tier, mode, streams and slices, Three Gates, codification, matrices, indices) and the
AI-use statement listing the tasks you performed in this project with the verification steps I
applied.
```

Your key: Compare every number with the calculator and the toolkit; sign the statement as the author.

MODEL AI-USE STATEMENT

"Generative AI (Claude, Anthropic; [month year]) was used for search-string ideation, first-pass extraction, provisional 6R scoring and drafting. All AI-assisted outputs were verified by the author(s) against the original sources in line with the 360° MDL Two-Key Rule. Indices and figures were computed with the 360° MDL Index Calculator (Lytras, 2026). The author(s) take full responsibility for the content."

CHAPTER 7

The complete workflow, tool by tool

Phase	You do	Claude may assist (Two-Key Rule)	Tool
P1 Frame	Purpose, questions, scope, tier, clusters, eligibility	Draft the Canvas; question your scope	Toolkit T1; Playbook Ch. 4
P2 Source	Run and log searches in five families	Search strings; source menus; candidate triage from pasted exports	Toolkit T3; Playbook Ch. 5–6
P3 Select	Gate 1 eligibility; Gate 2 6R scores; Gate 3 CCD	Provisional scores with rationale	Toolkit T4, T5; **Calculator: CCD checks, flow diagram**
P4 Codify	IDs, Master Table, keyword codebook	First-pass extraction; clustering proposals	Toolkit T6, M4
P5 Analyse	Matrices M1–M6	Cross-matrix pattern suggestions	Toolkit M1–M6; **Calculator: CovI, MDI**
P6 Synthesise	Key findings, triangulation, M7, M8	Draft findings with IDs; framework wording	Toolkit T15, M7, M8; **Calculator: TS, RI, LOSO, figures**
P7 Report	Methods, checklist, open data, AI statement	Methods draft from the report; statement draft	Toolkit T16, T17; **Calculator: checklist, methods sentence, report, JSON**

Frequently asked questions

Is my data safe? Yes. The page is a static file; it has no server. You can run it with the network switched off.

Can I use it with a toolkit from another language version? Yes, as long as the sheet names start with T1, T4, T6, M4 and T15 and the ID scheme is kept.

My CovI is low although my corpus is good. Check that the “x” marks in M4 are lower-case x in the cluster columns, and that cluster names are in row 4.

Why does LOSO show a value at the Lite tier? It is required only at Advanced, but reporting it at every tier tells readers which family your conclusions depend on.

Can I cite the calculator? Yes: Lytras, M. D. (2026). 360° MDL Index Calculator (beta 1) [Software]. Zenodo. https://doi.org/10.5281/zenodo.24033376

I found a bug or want a feature. Write to mlytras@acg.edu with the subject “360° MDL tool”, or deposit your suggestion in the Zenodo community. Beta 2 is planned to read the Stream Log (T3), offer a Greek interface and export vector (SVG) figures.

360° MDL

From volume to value

A method is only as useful as the tools that make it easy to apply. This calculator is offered freely to every researcher who wants a literature review that can be audited, trusted and built upon.

F1 F2 F3 F4 F5

Tool and guide: https://doi.org/10.5281/zenodo.24033376
Methodology: https://doi.org/10.5281/zenodo.23724393 · Playbook: https://doi.org/10.5281/zenodo.23777000
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